

# Determinant factorization

## Key Points

Row/column operations affect the determinant by:

- Adding/Subtracting multiples of other rows/columns and swapping rows/columns has no effect on determinant
- Multiplying rows/columns multiplies the determinant

Triangular matrices determinants are the product of the diagonal and hence turning a matrix into a diagonal can help to determine the determinant

## Basic Skills

**Q1.** Evaluate the following determinants

a)  $\begin{vmatrix} 6 & 2 \\ 1 & 4 \end{vmatrix}$     b)  $\begin{vmatrix} -5 & 0 \\ 1 & 2 \end{vmatrix}$     c)  $\begin{vmatrix} 6 & 0 & 0 \\ 4 & -5 & 0 \\ 3 & 2 & 1 \end{vmatrix}$

**Q2.** Given that  $|M| = 5$  and  $M'$  swaps two rows, state the determinant of  $M'$

**Q3.** Given that  $|M| = 5$  and  $M'$  is formed by multiplying a row of  $M$  by 2, state the determinant of  $M'$

**Q4.** Given that  $|M| = -4$  and  $M'$  is formed by multiplying two columns of  $M$  by 3, state the determinant of  $M'$

**Q5.** For a  $3 \times 3$  matrix  $M$  such that  $|M| = 2$  find  $|4M|$

## Competency Skills

**Q6.** State the constant  $k$  for the following equations

a)  $\begin{vmatrix} 6 & 2 \\ 1 & 4 \end{vmatrix} = k \begin{vmatrix} 12 & 4 \\ 1 & 4 \end{vmatrix}$     b)  $\begin{vmatrix} 6 & 2 \\ 1 & 4 \end{vmatrix} = k \begin{vmatrix} 3 & 1 \\ 1 & 4 \end{vmatrix}$     c)  $\begin{vmatrix} 6 & 2 \\ 1 & 4 \end{vmatrix} = k \begin{vmatrix} 3 & 1 \\ -3 & -12 \end{vmatrix}$     d)  $\begin{vmatrix} 10 & 3 \\ 2 & -1 \end{vmatrix} = k \begin{vmatrix} 5 & 9 \\ 6 & 6 \end{vmatrix}$

**Q7.** Show that  $\begin{vmatrix} 2 & 3 & -10 \\ 4 & 10 & 6 \\ 6 & 5 & 8 \end{vmatrix} = \begin{vmatrix} 2 & 3 & -10 \\ 0 & 4 & 26 \\ 0 & 0 & 64 \end{vmatrix}$  and hence state the value

**Q8.** Explain why  $5x + 16$  is a factor of  $\begin{vmatrix} x & 5 & 3 \\ x+2 & -2 & 5 \\ x & 0 & 8 \end{vmatrix}$

**Q9.** Express the following matrix determinants in factorized form

a)  $\begin{vmatrix} 2x-3 & 3 & 2 \\ 2x-2 & 3x+11 & 3 \\ 6x-1 & 6x-1 & 6x-1 \end{vmatrix}$     b)  $\begin{vmatrix} x-4 & 1 & 2 \\ x+3 & x-9 & 5 \\ x-6 & x & 2x \end{vmatrix}$

## Advanced Skills

**Q10.** Explain why there is no difference between the rules for row/column operations when considering the effect on the determinant

**Q11.** For  $M = \begin{pmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{pmatrix}$  show that  $|M| = abc(b-a)(c-a)(c-b)$  and hence state when the matrix  $M$  is singular

**Q12.** Factorize the following determinants

a)  $\begin{vmatrix} x^2 & xy & x \\ y^2 & xz & y \\ z^2 & zy & z \end{vmatrix}$     b)  $\begin{vmatrix} 1 & 1 & 1 \\ 1 & \cos x & \cos^2 x \\ 1 & \sin x & \sin^2 x \end{vmatrix}$

## Extension

**Q13.** Find when the matrix  $\begin{pmatrix} 1 & 1 & 1 \\ 1 & \sin x & \cos x \\ 1 & \sin^2 x & \cos^2 x \end{pmatrix}$  is singular